

NyayVault: Comprehensive System Dossier

AI-Powered Secure Digital Evidence Management & Case Intelligence System (Problem Statement: SIH26190)

Hackathon: Smart India Hackathon 2026

Problem Statement ID: SIH26190

Theme: Smart Automation / LegalTech / Cyber Security

Target Compliance: BSA 2023, BNSS 2023, IT Act

Tech Stack: React 18, Node.js, PostgreSQL (pgvector), Gemini Flash

Deployment: AWS EC2, Vercel Edge, MinIO S3

1. Executive Summary & Problem Context

Over 5 Crore pending cases in Indian courts are severely bottlenecked by manual, paper-based evidence management. Physical documents in police station *Malkhanas* (evidence rooms) are vulnerable to theft, moisture decay, fire, and illegal substitution. Furthermore, proving the authenticity and custody trail of electronic evidence under **Section 63 of the Bharatiya Sakshya Adhiniyam (BSA 2023)** is extremely difficult without an immutable cryptographic audit record. **NyayVault** is an end-to-end digital evidence ecosystem providing real-time SHA-256 cryptographic verification, zero-trust Role-Based Access Control (RBAC), automated AI OCR and PII Redaction, and case-bound RAG AI intelligence.

2. Zero-Trust Role-Based Access Control (RBAC)

The platform enforces strict legal and jurisdictional boundaries across 6 distinct roles:

Role Name	User Badge	Permissions & Legal Boundary
Investigating Officer (IO)	POL-1	Files citizen complaints, registers FIRs, uploads seizure/remand docs, initiates AI redactions. Can only view own investigation files until submitted to court.
Forensic Analyst	FOR-1	Uploads Cyber Forensics, Bank Tracing, DNA/Ballistic reports; computes cryptographic SHA-256 signatures.
Judicial Officer / Judge	JUD-1	Full unredacted access across all partitioned folders, queries AI Case Assistant, verifies cryptographic integrity, delivers judgments.
Defense Advocate	ADV-1	Uploads bail petitions (Sec 489 BNSS) and defense evidence; view restricted strictly to filed court records (blocked from confidential police diaries).
Prosecution Lawyer	ADV-2	Uploads bail objections and prosecution filings; accesses verified police and forensic reports.
Court Registrar	REG-1	Issues digital summons to witnesses/banks, maintains court trial schedule, manages official dockets.

3. Core Technical Modules & Feature Matrix

A. Cryptographic Chain of Custody & Live Tamper Engine

- SHA-256 Digital Fingerprint:** Instant hashing on upload; any bit-level modification triggers TAMPER_DETECTED warnings.
- BSA Section 63 Digital Certificates:** Auto-generates legal electronic evidence certificates with device IP, timestamp, and signature digest.
- Immutable Audit Ledger:** Append-only logging of uploads, downloads, views, and integrity checks.

B. AI Document Intelligence & OCR Extraction

- **Automated OCR:** Extracts raw text from scanned PDFs, PNGs, and handwritten legal forms via Google Gemini 2.5 Flash.
- **Legal Classification:** Classifies FIR, Charge Sheet, Remand Application, Bail Petition, Summons with confidence ratings.
- **Structured Metadata:** Automatically extracts key entities (Accused Name, FIR No, Police Station, Crime Sections, Dates).

C. AI Redaction Studio (DPDP Act Compliance)

- **PII Masking:** NLP & regex scan for Aadhaar numbers, PAN, bank accounts, contact details, and victim identities.
- **Non-Destructive Versioning:** Creates a sanitized clone for court record while keeping original locked in Judicial Vault.

D. Case Intelligence & RAG Judicial Copilot

- **Vector Search (pgvector):** Case documents converted to embeddings (gemini-embedding-001) for semantic querying.
- **Zero Hallucination:** System prompt strictly binds AI answers to citations from verified case files.
- **Executive 1-Page Summary:** Auto-synthesizes Case Overview, Chronological Timeline, Key Evidence, and Pending Actions.

E. Citizen Complaint to FIR Pipeline

- Digital pipeline for victim complaints (PENDING → FIR_FILED / REJECTED) with mandatory IO justification.

4. Technical Architecture & Database Schema

The system follows a multi-tier microservices-compatible architecture: **Frontend:** React 18, Vite, Tailwind CSS Dark UI, Framer Motion. **Backend:** Node.js, Express, JWT Auth, PM2 on AWS EC2 (3.82.17.79). **Database:** PostgreSQL with pgvector + embedded PGLite engine for instant resilience. **Storage:** S3-Compatible MinIO object store with local filesystem caching.

Database Table	Key Columns & Purpose
users	id, badge_number, full_name, role, department, password_hash — Multi-role credentials.
cases	id, case_number, title, description, status, created_by — Master legal dockets.
documents	id, case_id, filename, storage_key, sha256_hash, document_type, document_category, classification_confidence, metadata, extracted_text, is_redacted, parent_document_id — Evidence registry.
document_chunks	id, document_id, chunk_index, chunk_text, embedding VECTOR(768) — Semantic RAG vector index.
complaints	id, case_id, complainant_name, contact, complaint_text, status, io_remarks, rejection_reason — Victim pipeline.
audit_logs	id, case_id, document_id, user_id, action, ip_address, metadata, timestamp — Immutable chain of custody.

5. Live Demo Case Study: 'State vs Hardik' (■8.5L Cyber Fraud)

Pre-seeded with 11 authentic legal evidence documents covering the entire trial:

1. 01_Initial_Complaint.png — Victim Rahul Sharma complaint (■8.5L fraud).
2. 02_FIR_Copy.png — FIR 112/2026 u/s 318(4) BNS & 66D IT Act.
3. 03_Arrest_Memo.png — Accused Hardik Verma arrested; laptop seized.
4. 04_Remand_Application.png — IO requests 3-day custody for wallet keys.
5. 05_Cyber_Forensics_Report.png — FSL report linking laptop IP to crypto wallet.
6. 06_Defense_Bail_Application_Hardik.png — Bail plea u/s 489 BNSS.
7. 07_Prosecution_Bail_Objection.png — State objection on flight risk.
8. 08_Final_Charge_Sheet_Sec173.png — Police Charge Sheet u/s 173 CrPC / 193 BNSS.
9. 09_Court_Summons_Bank_Manager.png — Witness summons to ICICI Bank Manager.

10. 10_Official_Trial_Schedule.png — Official trial calendar and hearing dates.

11. 11_Final_Court_Judgment.png — Conviction under Sec 318(4) BNS (3 yrs RI + ■10L fine).

6. Statutory Alignment Matrix

Statutory Act	Section / Mandate	NyayVault Implementation
BSA 2023	Sec 61 & 63	Automated SHA-256 electronic certificates ensuring legal admissibility of digital records.
BNSS 2023	Sec 173/193 & 489	Digital workflow for charge sheets, police diaries, and bail hearing filings.
IT Act 2000	Sec 65B & 66D	Immutable device IP, badge ID, and cryptographic timestamp logging for cyber evidence.
DPDP Act 2023	Data Privacy & Minimization	AI Redaction Studio sanitizes Aadhaar, PAN, and victim identities non-destructively.

7. How to Feed this Dossier to Any LLM for PPT Generation

Copy and paste this document into ChatGPT, Claude, Gemini, Gamma.app, or SlidesAI with the prompt:

"You are an expert pitch deck designer. Based on this complete NyayVault Project Dossier (SIH26190), generate a 10-slide Smart India Hackathon presentation with slide titles, bullet points, architecture diagrams, and 30-second speaker scripts."